Project – Hotel Booking Analysis

Introduction:

This Power BI project focuses on analyzing hotel booking data to derive actionable insights into room occupancy, booking trends, cancellation patterns, revenue performance, and customer behavior. By leveraging DAX measures, calendar-based time intelligence, and interactive dashboards, this solution enables stakeholders to explore hotel performance across different dimensions including room types, market segments, booking channels, and geographic regions. The implementation also incorporates dynamic metric and period selectors, making the analysis adaptable for decision-making across operational, financial, and marketing functions.

Steps:

1. **Imported Data**
   - Loaded hotel data from pre-processed CSV format.
   - Verified row counts for data integrity.
2. **Checked Data Quality**
   - Opened Power Query Editor.
   - Enabled Column Quality, Distribution, Profile.
   - Ensured:
     - 100% valid values.
     - 0% errors.
   - Transformed data:
     - Converted Arrival Date format.
     - Created calculated fields

```
Weekend Rev = 'hotel data'[Stays In Weekend Nights] * 'hotel data'[Adr]
Week Nights Revenue = 'hotel data'[Stays In Week Nights] * 'hotel data'[Adr]
```

3. **Created Date Table**
   - Created a new table

```
Date Table = CALENDAR(MIN('hotel data'[Arrival Date]), MAX('hotel data'[Arrival Date]))
```

   - Added below columns:

```
Month = FORMAT('Date Table'[Date], "mmm")
Month Number = MONTH('Date Table'[Date])
Year = YEAR('Date Table'[Date])
Week = WEEKNUM('Date Table'[Date])
Quarter = QUARTER('Date Table'[Date])
Quarter Name = CONCATENATE("Q" , 'Date Table'[Quarter])
```

   - Created relationship with `'hotel data'` to 'Date Table' (many-to-many).

     [Date Table[Date] <–> hotel data[Arrival Date]]

4. **Created Measures**
   - **Total Countrows and Sum Measures:**

```
Total Rooms Booked = CALCULATE(COUNTROWS('hotel data'))
Total Occupied Rooms = CALCULATE(COUNTROWS('hotel data'), 'hotel data'[Is Canceled] =0)
Total Days of Booking = CALCULATE(SUM('hotel data'[Total days]), 'hotel data'[Is Canceled] = 0)
Total Weekend Nights = CALCULATE(SUM('hotel data'[Stays In Weekend Nights]), 'hotel data'[Is Canceled]=0)
```

```
Total Weekday Nights = CALCULATE(SUM('hotel data'[Stays In Week Nights]), 'hotel data'[Is Canceled]=0)
Total Cancelled Bookings = CALCULATE(COUNTROWS('hotel data'), 'hotel data'[Is Canceled]=1)
Total Revenue = CALCULATE(SUM('hotel data'[Rev]), 'hotel data'[Is Canceled]=0)
Week Nights Rev = CALCULATE(SUM('hotel data'[Week Nights Revenue]), 'hotel data'[Is Canceled] =0)
Weekend Nights Rev = CALCULATE(SUM('hotel data'[Weekend Nights Revenue]), 'hotel data'[Is Canceled] =0)
Sum of Occupancy for each Week = SUMX(VALUES('Date Table'[Week]), CALCULATE([Total Occupied Rooms]))
Sum of Total Rev for each Week = SUMX(VALUES('Date Table'[Week]), CALCULATE([Total Revenue]))
```

- **Average Measures:**

```
Occupancy Rate = [Total Occupied Rooms] / [Total Rooms Booked]
Avg Daily Rate (ADR) = CALCULATE(AVERAGE('hotel data'[Adr]))
Avg Days of Stay = AVERAGE('hotel data'[Total days])
Avg Week Occupied Rooms = CALCULATE(AVERAGEX(VALUES('Date Table'[Week]), CALCULATE([Total Occupied Rooms])), ALL('Date Table'))
Avg Week Revenue = CALCULATE(AVERAGEX(VALUES('Date Table'[Week]), CALCULATE(SUM('hotel data'[Rev]))), ALL('Date Table'))
Sum of Avg Sales each week = SUMX(VALUES('Date Table'[Week]), [Avg Week Revenue])
Rev PAR = [Average Daily Rate (ADR)] * [Occupancy Rate]
```

5. **Created table for Selecting Metric:**

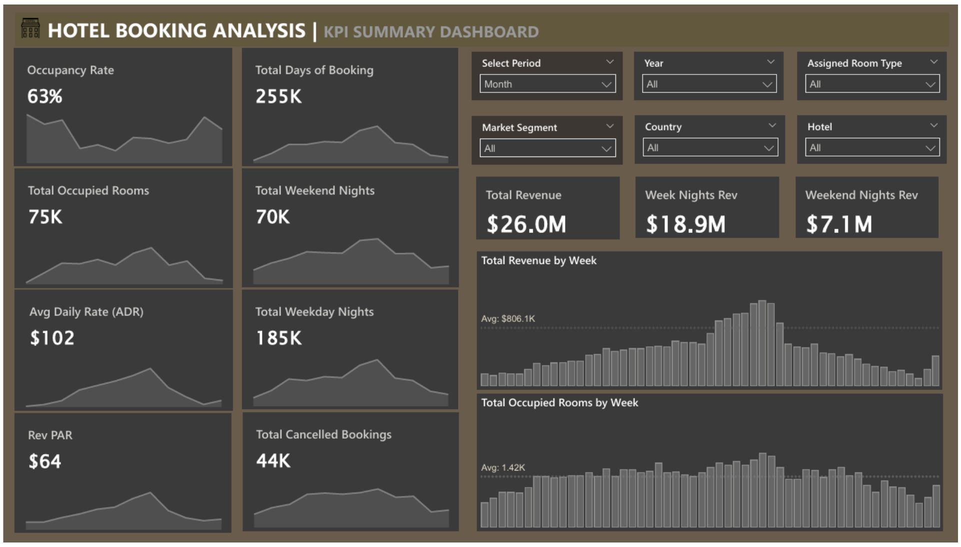
```
Select Metric = {
    ("Total Revenue", NAMEOF('hotel data'[Total Revenue]), 0),
    ("Total Days of Booking", NAMEOF('hotel data'[Total Days of Booking]), 1),
    ("Total Occupied Rooms", NAMEOF('hotel data'[Total Occupied Rooms]), 2),
    ("Total Weekday Nights", NAMEOF('hotel data'[Total Weekday Nights]), 3),
    ("Total Weekend Nights", NAMEOF('hotel data'[Total Weekend Nights]), 4)
}
```

6. **Created table for Selecting Period:**

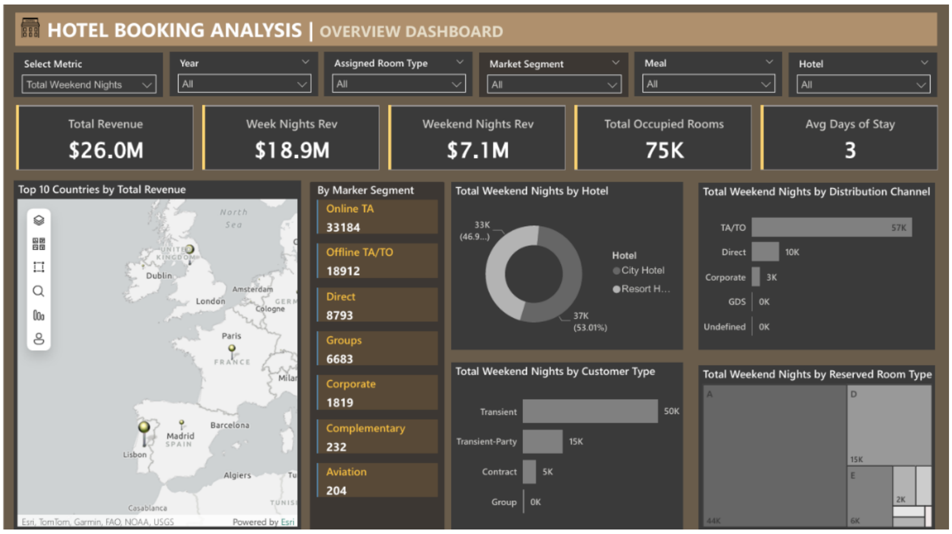
```
Select Period = {
    ("Month", NAMEOF('Date Table'[Month]), 0),
    ("Quarter", NAMEOF('Date Table'[Quarter Name]), 1),
    ("Week", NAMEOF('Date Table'[Week]), 2)
}
```

7. **Created KPI Summary Dashboard**
    - Added Dashboard Title
    - Added Slicers
        - Select Period, Year, Assigned Room Type, Market Segment, Country, and Hotel
    - Added Card to show Occupancy Rate as percentage and Area Chart to show data over the Period Type (Week/Qtr/Month)
    - Added Card to show Total Days of Booking (excluding the canceled ones) and Area Chart to show data over the Period Type (Week/Qtr/Month)
    - Added Card to show Total Occupied Room (excluding the canceled ones) and Area Chart to show data over the Period Type (Week/Qtr/Month)

- Added Card to show Total Days of Booking on Weekend Nights (excluding the canceled ones) and Area Chart to show data over the Period Type (Week/Qtr/Month)
- Added Card to show Average Daily Rate and Area Chart to show data over the Period Type (Week/Qtr/Month)
- Added Card to show Total Days of Booking on Weekday Nights (excluding the canceled ones) and Area Chart to show data over the Period Type (Week/Qtr/Month)
- Added Card to show Revenue Per Available Room and Area Chart to show data over the Period Type (Week/Qtr/Month)
- Added Card to show Total Days of Booking Canceled and Area Chart to show data over the Period Type (Week/Qtr/Month)
- Added 3 Data Cards to show Total Revenue, Weeknights Revenue and Weekday Revenue
- Added a Stacked Column Chart to show Total Revenue by Week
- Added a Stacked Column Chart to show Total Occupied Room by Week



8. **Created Overview Dashboard**
   - Added Dashboard Title
   - Added Slicers
     - Select Metric, Year, Assigned Room Type, Market Segment, Meal, and Hotel
   - Added Card to show:
     - Total Revenue, Week Nights Revenue, Weekend Nights Revenue, Total Occupied Rooms, and Average Days of Stay
   - Added an ArcGIS Map to show Top 10 Countries by Total Revenue
   - Added a Multirow Card to show data by Market Segment
   - Added a Pie Chart to show ratio by Hotel based on Selected Metric
   - Added a Stacked Bar Chart to show data by Distribution Channel based on Selected Metric
   - Added a Stacked Bar Chart to show data by Customer Type based on Selected Metric
   - Added a Treemap to show proportion by Reserved Room Type based on Selected Matric



9. **Dashboard Design and Formatting**
    - Dark theme background
    - Chocolate color palette
    - Rounded corners and visual shadows
    - Consistent font and title formatting

Key Findings:

1. **Key KPIs**
    - **Occupancy Rate:** 63% - indicates moderate room utilization.
    - **Total Occupied Rooms:** 75K - reflects the volume of bookings during the period.
    - **Avg Daily Rate (ADR):** $102 - average revenue per occupied room per day.
    - **RevPAR:** $64 - revenue per available room, combining occupancy and ADR.
    - **Total Cancelled Bookings:** 44K - highlights a significant number of cancellations, which may impact revenue.
2. **Revenue Breakdown**
    - **Total Revenue:** $26.0M - overall revenue generated.
    - **Week Nights Revenue:** $18.9M (72.7% of total) - majority of revenue comes from weekday stays.
    - **Weekend Nights Revenue:** $7.1M (27.3% of total) - weekends contribute less but still significant.
3. **Booking Trends**
    - **Total Weekend Nights:** 70K vs. **Total Weekday Nights:** 185K - weekday stays dominate bookings.
    - **Avg Days of Stay:** 3 - suggests most guests book short-term stays.
4. **Top Markets**
    - **Top 10 Countries by Revenue:** Includes North Sea, Dublin, London, Paris, and others - European destinations are major revenue drivers.
    - **Market Segment Distribution:**
        - **Online TA (Travel Agent):** 33,184 bookings - the largest segment.
        - **Offline TA/TO:** 18,912 - significant but smaller than online.
        - **Direct Bookings:** 8,793 - indicates room for growth in direct customer engagement.
5. **Room and Hotel Insights**
    - **Total Weekend Nights by Hotel:**
      City Hotel 37K (dominant) vs. **Resort Hotel** - city hotels attract more weekend stays.

- **Reserved Room Types:** Room types A and D are most common, with transient parties and contracts also notable.

6. **Cancellations and Operational Metrics**
   Cancellation Rate: ~37% (44K cancellations out of ~119K total bookings) - high cancellation rate may require strategy adjustments.
7. **Insights:**
   - Revenue is heavily weighted toward weekday stays, suggesting potential to optimize weekend promotions.
   - Online travel agents are the primary booking channel, while direct bookings lag, indicating an opportunity to improve direct sales strategies.
   - High cancellation rates and moderate occupancy suggest a need for better retention or cancellation policies.
   - European markets dominate revenue, highlighting their importance for targeted marketing efforts.

Conclusion:
The hotel booking analysis dashboards created in Power BI provide a comprehensive and interactive view of key performance indicators such as occupancy rate, total revenue, and booking distribution over time. With dynamic selectors and well-structured visuals, users can easily explore trends, compare performance across periods, and identify areas for strategic improvements. The combination of clean design, thoughtful metric calculations, and responsive filtering ensures a data-driven approach to understanding guest behavior, optimizing revenue, and enhancing operational efficiency.